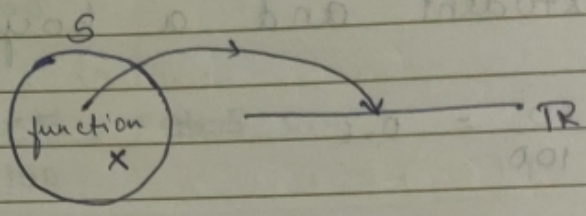
DATE: PAGE:

Random Variables.

A random variable X is a function which maps the elements of sample space S to the set of real numbers $\mathbb{R}$ i.e., the S of the random exp. if a real no. is associated with every outcome then it is called a random variable. (i.e., $X : S \to \mathbb{R}$)



We use the upper case letters for random variables

Note

1) While tossing a coin, suppose that the value 1 is associated for the outcome Head and 0 for the outcome Tail. We have $S = \{H, T\}$ and if X is a random variable then $X(H) = 1$ and $X(T) = 0$. Therefore the range of the random variable $X = \{0, 1\}$

Consider the exp. of tossing a coin 2 times. For this exp. the sample space $S = \{HH, HT, TH, TT\}$. Let X be the no. of heads in the outcome, then $X(HH) = 2$, $X(HT) = 1$, $X(TH) = 1$, $X(TT) = 0$. $\therefore X = \{0, 1, 2\}$

DATE: PAGE:

Discrete Random Variable

If a random variable takes finite or countably infinite no. of values then it is called a discrete random variable

Eg:
* The no. of students in a class
* The results of rolling two dice.

Continuous Random Variable

If a random variable takes infinite / uncountable no. of values then it is called continuous random variable.

eg:
* A person's height (could be any value within the range of human heights not just certain fixed hights
* Time in a race. (you could even measure it to the fraction of seconds).
* Weight of articles.

Discrete Probability Distribution

If For each value x_i of a discrete random variable (x) we assign a real number $p(x_i)$ such that (i) $p(x_i) \geq 0$ (ii) $\Sigma p(x_i) = 1$ then the function $p(x)$ is called a probability function

If the probability that x take the values x_i is p_i then (probability of $x = x_i$) $[p(x = x_i)]$ $= p_i$ or $P(x_i)$

The set of values $[x_i, p(x_i)]$ is called a discrete probability distibution of the discrete random variable x. the function $P(x)$ is

DATE: PAGE:

called the probability density function (PDF) or probability mass function (PMF).
The distribution function $f(x)$ defined by $f(x) = P(x \leq x) = \sum_{i=1}^{x} P(x_i)$ (where x being an integer) is called the cumulative distribution function.
The mean and variance of the discrete probability distribution are defined as follows.

Mean $= \mu = E(x) = \sum x_i p(x_i)$.
↳ expectation

Variance $= V = \sigma_x^2 = \sum (x_i - \mu)^2 P(x_i)$
$= \sum x_i^2 P(x_i) - \mu^2$ (most used)
$= E(x^2) - \mu^2$

Problems.

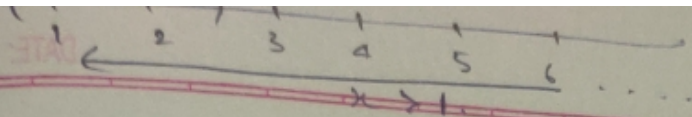
1. If X is a discrete random variable having $p(x)$ and is defined as $p(x) = \begin{cases} \frac{x}{15} & \text{if } 1 \leq x \leq 5 \\ 0 & \text{if } x > 5 \end{cases}$

S.T. $p(x)$ is a probability mass function and also find

i) $P(x = 1 \text{ or } 2)$

ii) $P\left(\frac{1}{2} < x < \frac{5}{2} \mid x > 1\right)$

=> The probability distribution is $x: 1\ 2\ 3\ 4\ 5\ 6$



DATE: PAGE:

x:	1	2	3	4	5	6	7	.....
$p(x)$:	$\frac{1}{15}$	$\frac{2}{15}$	$\frac{3}{15}$	$\frac{4}{15}$	$\frac{5}{15}$	0	0	.....

Here $p(x) \geq 0$ for each x, then.

$$\Sigma p(x) = \frac{1}{15} + \frac{2}{15} + \frac{3}{15} + \frac{4}{15} + \frac{5}{15}$$

$$= \frac{15}{15} = 1.$$

$\therefore \Sigma p(x) = 1$.

Hence $p(x)$ is a probability density/mass function

i) $P(X=1 \text{ or } X=2) = P(X=1) \text{ or } (+) P(X=2)$

$$= \frac{1}{15} + \frac{2}{15}$$

$$= \frac{3}{15} = \frac{1}{5}$$

ii) $P\left(\frac{1}{2} < x < \frac{5}{2} \mid x \geq 1\right) = \frac{P\left[\left(\frac{1}{2} < x < \frac{5}{2}\right) \cap (x \geq 1)\right]}{P(x \geq 1)}$.

$$= \frac{P[X=2]}{1 - P(X \leq 1)}$$

$$= \frac{2/15}{1 - P(X=1)} = \frac{2/15}{1 - 1/15} = \frac{1}{7}.$$

DATE: PAGE:

4. A random variable X has $P(x) = 2^{-x}$ where $x = 1, 2, 3, \ldots$. ST $P(x)$ is a probability function also find

i) $P(x$ is even$)$

ii) $P(x$ being divisible by $3)$

iii) $P(x \geq 5)$

=> Clearly $p(x) > 0$ for each x.

Now we need to P.T $\sum p(x) = 1$

$$\sum p(x) = \sum_{x=1}^{\infty} 2^{-x}$$

$$= 2^{-1} + 2^{-2} + 2^{-3} + \ldots$$

$$= \frac{1}{2}\left[1 + \frac{1}{2} + \left(\frac{1}{2}\right)^2 + \ldots\right]$$

$$= \frac{1}{2}\left[\frac{1}{1 - 1/2}\right] = 1.$$

$\therefore p(x)$ is a P.F.

i) $P[x$ is even$] = P[x=2] + P[x=4] + P[x=6] + \ldots$

$$= 2^{-2} + 2^{-4} + 2^{-6} + \ldots$$

$$= \frac{1}{2^2}\left[1 + \left(\frac{1}{2^2}\right) + \left(\frac{1}{2^3}\right)^2 + \ldots\right]$$

$$= \frac{1}{2^2}\left[\frac{1}{1 - 1/2^2}\right]$$

$$= \frac{1}{3}$$

DATE: PAGE:

ii) $p[x \text{ being divisible by } 3]$

$= p(x=3) + p(x=6) + p(x=9) + \ldots$

$= 2^{-3} + 2^{-6} + 2^{-9} + \ldots$

$= \frac{1}{2^3}\left[1 + \left(\frac{1}{2^3}\right) + \left(\frac{1}{2^3}\right)^2 + \ldots\right]$

$= \frac{1}{8} \times \left[\frac{1}{1 - 1/2^3}\right] = \frac{1}{7}.$

iii) $P(x \geq 5) = P(x=5) + P(x=6) + P(x=7) + \ldots$

$= 2^{-5} + 2^{-6} + 2^{-7}$

$= 1/2^5 [1 + (1/2) + (1/2)^2 + \ldots]$

$= 1/2^5 \times 1/1 - 1/2 \Rightarrow 1/16$

5. A random variable X takes the values $-3, -2, -1, 0, 1, 2, 3$ such that $P(x=0) = P(x<0)$ and $P(x=-3) = P(x=-2) = P(x=-1) = P(x=1) = P(x=2) = P(x=3)$. Find the P.Distribution.

$\Rightarrow$ Let the distribution $(X, P(x))$ be as follows.

X	-3	-2	-1	0	1	2	3
$P(x)$	p_1	p_2	p_3	p_4	p_5	p_6	p_7

Since, $P(x=0) = P(x<0)$.

$P(x=0) = P(x=-3) + P(x=-2) + P(x=-1)$

$p_4 = p_1 + p_2 + p_3 \longrightarrow (1)$

$P(x=-3) = P(x=-2) = P(x=-1) = P(x=1) = P(x=2) = P(x=3)$

$p_1 = p_2 = p_3 = p_5 = p_6 = p_7 \longrightarrow (2)$

Since $\sum P(x) = 1$.

$P(x=-3) + P(x=-2) + P(x=-1) + P(x=0) + P(x=1) + P(x=2) + P(x=3) = 1$

$p_1 + p_2 + p_3 + p_4 + p_5 + p_6 + p_7 = 1 \rightarrow (3)$.

Now eqn (1) becomes:

$p_4 + p_1 + p_1 + p_1 \Rightarrow p_4 = 3p_1$

Now eqn (3) becomes.

$p_1 + p_1 + p_1 + p_4 + p_1 + p_1 + p_1 = 1.$

$6p_1 + 3p_1 = 1$

$9p_1 = 1$

$p_1 = \frac{1}{9}$; $p_4 = \frac{3}{9}$

Thus P.D is

6. If the random variable X take the values 1, 2, 3, 4 such that $2P(x=1) = 3P(x=2) = P(x=3) = 5P(x=4)$. Find the probability distribution function of X.

$\Rightarrow$ Let the distribution $[X, P(X)]$ as follows

X	:	1	2	3	4
P(X)	:	p_1	p_2	p_3	p_4

Given that $2P(x=1) = 3P(x=2) = P(x=3) = 5P(x=4)$

$2p_1 = 3p_2 = p_3 = 5p_4$.

DATE: PAGE:

$p_2 = \frac{2}{3} p_1$

$p_3 = 2p_1$

$p_4 = \frac{2}{5} p_1$.

Since $\Sigma P(x) = 1$.

$P(x=1) + P(x=2) + P(x=3) + P(x=4) = 1$.

DATE: PAGE:

7. From a seed box containing a dozen Apples it was found that 3 Apples are perished. Obtain the probability distribution of the no. of perished Apples when two Apples are drawn at random, also find the mean and variance of this distribution.

=> Let X be the no of perished Apples. Since 2 Apples are drawn at random, then X is $= \{0, 1, 2\}$

$P(X=0 \text{ i.e. no perished apples}) = \frac{9C_2}{12C_2} = \frac{6}{11}$

$P(X=1) = \frac{3C_1 \times 9C_1}{12C_2} = \frac{9}{22}$

$P(X=2) = \frac{3C_2}{12C_2} = \frac{1}{22}$

The P D is

X :	0	1	2
P(x) :	$\frac{9C_2}{12C_2}$	$\frac{3C_1 \times 9C_1}{12C_2}$	$\frac{3C_2}{12C_2}$

Mean $= \mu = \sum X P(X)$

$= \left[0 \times \frac{9C_2}{12C_2}\right] + \left[1 \times \frac{3C_1 \times 9C_1}{12C_2}\right] + \left[2 \times \frac{3C_2}{12C_2}\right]$

$= 0.5 = \frac{1}{2}$

DATE: PAGE:

Variance, $V = \Sigma x^2 P(x) - \mu^2$

$= \left[0 \times \frac{6}{11}\right] + \left[1^2 \times \frac{9}{22}\right] + \left[2^2 \times \frac{1}{22}\right] - \left(\frac{1}{2}\right)^2$

$= \frac{15}{44}$ //

Q A coin is tossed twice. A random variable X represent the no of X turning up. Find the discrete probability for X and also find its mean and variance. ~~The~~

=> The sample space of tossing a coin twice is $S = \{HH, HT, TH, TT\}$

Since X is the no of heads turning up then $X = \{0, 1, 2\}$

$P(X=0) = \frac{1}{4}$

$P(X=1) = \frac{2}{4}$

$P(X=2) = \frac{1}{4}$

∴ The discrete probability distibution for X is

X	0	1	2
P(x)	1/4	2/4	1/4

Mean, $\mu = \Sigma x P(x)$

$= (0 \times 1/4) + (1 \times 2/4) + (2 \times 1/4) = \frac{4}{4} = 1$

DATE: PAGE:

Variance, $V = \sum x^2 P(x) - \mu^2$

$= 0^2 \times \frac{1}{4} + 1^2 \times \frac{2}{4} + 2^2 \times \frac{1}{4} - 1^2$

$= \frac{1}{2}$

9. A random exp. of tossing a die twice is performed. A random variable X on the sample space is defined to be the sum of two numbers turning up on the toss. Find the discrete probability distribution for the random variable X and compute the corresponding mean and S.D.

$\Rightarrow$ The sample space of tossing a die twice, we get $S = \{(1,1), \ldots, (6,6)\}$.

$|S| = 36$ [(Cardinality) = No. of elements in S]

Since X be the sum of 2 numbers turning up on the toss, then $X = \{2, 3, 4, 5, \ldots 12\}$

$P(X=2) = \frac{1}{36}$

$P(X=3) = \frac{2}{36}$ [$\because (2,1)(1,2)$]

$P(X=4) = \frac{3}{36}$ [$\because (2,2)(1,3)(3,1)$]

$P(X=5) = \frac{4}{36}$ [$\because (1,4)(4,1)(3,2)(2,3)$]

$P(X=6) = \frac{5}{36}$ [$\because (1,5)(5,1)(4,2)(2,4)(3,3)$]

$P(X=7) = \frac{6}{36}$ "

DATE: PAGE:

$P(X=8) = \frac{5}{36}$ "

$P(X=9) = \frac{4}{36}$ "

$P(X=10) = \frac{3}{36}$ "

$P(X=11) = \frac{2}{36}$ "

$P(X=12) = \frac{1}{36}$ "

∴ the Discrete probability distibution for X is

X :	2	3	4	5	6	7	8	9	10	11	12
P(x) :	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

Mean = $\mu = \Sigma x P(x) = \left(2 \times \frac{1}{36}\right) + \left(3 \times \frac{2}{36}\right) + \left(4 \times \frac{3}{36}\right) +$

$\left(5 \times \frac{4}{36}\right) + \left(6 \times \frac{5}{36}\right) + \left(7 \times \frac{6}{36}\right) + \left(8 \times \frac{5}{36}\right) + \left(9 \times \frac{4}{36}\right) + \left(10 \times \frac{3}{36}\right) +$

$\left(11 \times \frac{2}{36}\right) + \left(12 \times \frac{1}{36}\right) = 7$

Variance = $V = \Sigma x^2 P(x) - \mu^2$

$= \frac{35}{6}$.

S.D = $\sqrt{\frac{35}{6}}$ =

DATE: PAGE:

10. Find the value of k such that the following distribution represents a finite probability distribution. Hence find its μ and SD also find $P(x \leq 1)$ and $P(-1 < x < 2)$

x :	-3	-2	-1	0	1	2	3
$p(x)$:	k	$2k$	$3k$	$4k$	$3k$	$2k$	k

=> Since $p(x)$ is a probability distribution, then we have each $p(x) \geq 0$ and $\Sigma p(x) = 1$.

Then $k + 2k + 3k + 4k + 3k + 2k + k = 1$

$k = \frac{1}{16}$.

The distribution of $(x, p(x))$ is

x :	-3	-2	-1	0	1	2	3
$p(x)$:	$\frac{1}{16}$	$\frac{2}{16}$	$\frac{3}{16}$	$\frac{4}{16}$	$\frac{3}{16}$	$\frac{2}{16}$	$\frac{1}{16}$

Mean, $\mu = \Sigma x\, p(x)$

$= \left(-3 \times \frac{1}{16}\right) + \left(-2 \times \frac{2}{16}\right) + \left(-1 \times \frac{3}{16}\right) + 0 + \left(1 \times \frac{3}{16}\right) + \left(\frac{2}{16} \times 2\right) + \left(\frac{1}{16} \times 3\right) = 0.$

Variance, $V = \Sigma x^2 p(x) - \mu^2$

$= \frac{40}{16}$.

$S.D = \sqrt{V} = \sqrt{\frac{40}{16}} =$

DATE: PAGE:

$P(x \leq 1) = 1 - P(x > 1)$

$= 1 - [P(x=2) + P(x=3)]$

$= 1 - \left(\frac{2}{16} + \frac{1}{16}\right)$

$= \frac{13}{16}$.

$P(x > 1) = P(x=2) + P(x=3) = \frac{3}{16}$.

$P(-1 < x < 2) = P(x=0) + P(x=1) + P(x=2)$

$= 9/16$ //

11. A random variable x has the following probability function for various values of x as

x :	0	1	2	3	4	5	6	7
p(x) :	0	k	2k	2k	3k	k^2	$2k^2$	$7k^2+k$

(i) Find k (ii) Evaluate $P(x<6)$, $P(x \geq 6)$ and $P(3 < x \leq 6)$

⇒ Since P(x) is a probability distibution. then we have $\sum p(x=i)$ and each $p(x) \geq 0$. then.

$0 + k + 2k + 2k + 3k + k^2 + 2k^2 + 7k^2 + k = 1$.

$9k + 10k^2 - 1 = 0$.

$10k^2 + 9k - 1 = 0$.

$10k^2 + 10k - k - 1 = 0$.

$10k(k+1) - 1(k+1) = 0$.

$k = -1$ or $\frac{1}{10}$.

Hence $k = 1/10$ (∵ each $p(x) \geq 0$)

DATE: PAGE:

The distribution $(x, p(x))$ can be written as.

x	0	1	2	3	4	5	6	7
$p(x)$	0	$\frac{1}{10}$	$\frac{2}{10}$	$\frac{3}{10}$	$\frac{3}{10}$	$\frac{1}{100}$	$\frac{2}{100}$	$\frac{7}{100} + \frac{1}{10}$

$P(x<6) = 1 - P(x \geq 6)$.
$= 1 - [P(x=6) + P(x=7)]$ Ans =
$P(x \geq 6) = P(x=6) + P(x=7)$ Ans =
$P(3 < x \leq 6) = P(x=4) + P(x=5) + P(x=6)$ Ans =

Continuous Probability Distribution

If for every x belonging to the range of a continuous random variable X, we assign a real number $f(x)$ satisfying the conditions (i) $f(x) \geq 0$ (ii) $\int_{-\infty}^{\infty} f(x)dx = 1$, then

$f(x)$ is called a continuous probability function / probability density function.

Note

* $P[a \leq x \leq b] = P[a < x < b] = P[a < x \leq b] = P[a \leq x < b] = \int_a^b f(x)dx$.

If X is a continuous random variable with probability density function $f(x)$ then the $F(x)$ defined by $F(x) = P[X \leq x] = \int_{-\infty}^{x} f(x)dx$

is called the cumulative distribution function

of X.

$F(x) = P(X \leq x) = P[-\infty < X \leq x]$ and $\frac{d}{dx}[F(x)] = f(x)$.

If r is any real number then $P[x > r] = \int_r^\infty f(x)\,dx$. and $P[x < r] = 1 - P[x \geq r]$.

$= 1 - \int_r^\infty f(x)\,dx$.

Mean, $\mu = \int_{-\infty}^{\infty} x f(x)\,dx$.

Variance, $V = \sigma_x^2 = \int_{-\infty}^{\infty} (x-\mu)^2 f(x)\,dx$.

$= \int_{-\infty}^{\infty} x^2 f(x)\,dx - (\mu)^2$.

Problems

1. Let $f(x) = \begin{cases} 2x, & 0 < x < 1 \\ 0, & \text{otherwise.} \end{cases}$

Is $f(x)$ a P.d.f?

$\Rightarrow$ Clearly $f(x) > 0$ in $(-\infty, \infty)$

$\int_{-\infty}^{\infty} f(x)\,dx = \int_{-\infty}^{0} f(x)\,dx + \int_0^1 f(x)\,dx + \int_1^{\infty} f(x)\,dx$.

$$= \int_0^1 2x\,dx$$

$$= [x^2]_0^1 = 1$$

2. Let $f(x) = \begin{cases} 2x & -1 < x < 1 \\ 0 & \text{otherwise} \end{cases}$

Is $f(x)$ a p.d.f?

$\Rightarrow$ Clearly $f(x) \geq 0$ in $(-\infty, \infty)$

$$\int_{-\infty}^{\infty} f(x)\,dx = \int_{-\infty}^{-1} f(x)\,dx + \int_{-1}^{1} f(x)\,dx + \int_{1}^{\infty} f(x)\,dx.$$

$$= \int_{-1}^{1} 2x\,dx.$$

$$= [x^2]_{-1}^{1} = 1 - 1 = 0$$

$\therefore f(x)$ is not a pdf

3. Find the value of c such that

$$f(x) = \begin{cases} \frac{x}{6} + c, & 0 \leq x \leq 3 \\ 0, & \text{otherwise} \end{cases}$$

is a p.d.f

Also find $P[1 \leq x \leq 2]$

$\Rightarrow$ Given: $f(x)$ is a p.d.f.

$$= \int_{-\infty}^{\infty} f(x)\,dx = 1$$

$$= \int_{-\infty}^{0} f(x)\,dx + \int_{0}^{3} f(x)\,dx + \int_{3}^{\infty} f(x)\,dx$$

DATE: PAGE:

$$\int_0^3 \left[\frac{x}{6} + c\right] dx = 1$$

$$\left[\frac{x^2}{12} + cx\right]_0^3 = 1$$

$$\frac{9}{12} + 3c = 1 \Rightarrow c = \frac{1}{12} //$$

$$P[1 \le x \le 2] = \int_1^2 f(x)\,dx$$

$$= \int_1^2 \left(\frac{x}{6} + c\right) dx$$

$$= \left(\frac{x^2}{12} + cx\right)_1^2$$

$$= 1/3 + 2/12 + 1/12 - 1/12 = 1/3$$

4. Find the constant k such that

$$f(x) = \begin{cases} kx^2, & 0 < x < 3 \\ 0, & \text{otherwise} \end{cases}$$

is a P.d.f. Also compute

i) $P[1 < x < 2]$
ii) $P[x \le 1]$
iii) $P[x > 1]$
iv) Mean
v) Variance

Note

* If u and v are functions of x then $\int u.v\,dx = uv_1 - u'v_2 + u''v_3 - \ldots$ where $v_1 = \int v\,dx$, $v_2 = \int v_1\,dx$, $v_3 = \int v_2\,dx$, ... $u' = \frac{du}{dx}$, $u'' = \frac{du'}{dx}$, ...

$\Rightarrow$ Given that $f(x)$ is a P.d.f. then.

$$\int_{-\infty}^{\infty} f(x)\,dx = 1.$$

$$\int_{-\infty}^{0} f(x)\,dx + \int_{0}^{3} f(x)\,dx + \int_{3}^{\infty} f(x)\,dx = 1.$$

$$\int_{0}^{3} kx^2\,dx = 1.$$

$$k\left[\frac{x^3}{3}\right]_0^3 = 1. \Rightarrow k = \frac{1}{9}$$ //

i) $P[1 < x < 2] = \int_1^2 f(x)\,dx.$

$$= \int_1^2 kx^2\,dx.$$

$$= \frac{1}{27}\int_1^2 x^2\,dx.$$

$$\Rightarrow \frac{7}{27}$$ //.

ii) $P[x \le 1] = \int_{-\infty}^{1} f(x)\,dx = \int_{-\infty}^{0} f(x)\,dx + \int_0^1 f(x).dx.$

$$= \int_0^1 kx^2\,dx.$$

$$= \frac{1}{9}\left[\frac{x^3}{3}\right]_0^1 = \frac{1}{27}$$

$=$.

iii) $P[x > 1] = 1 - P(x \le 1)$

$= 1 - 1/27 = 26/27$ //

DATE PAGE

ii) Mean, $\mu = \int_{-\infty}^{\infty} x f(x)\,dx$.

$\mu = \int_{-\infty}^{0} x f(x)\,dx + \int_{0}^{3} x f(x)\,dx + \int_{3}^{\infty} x f(x)\,dx$

$\mu = \int_{0}^{3} x f(x)\,dx$.

$= k \int_{0}^{3} x \cdot x^2\,dx$

$= k \int_{0}^{3} x^3\,dx$.

$= \frac{1}{9}\left[\frac{x^4}{4}\right]_0^3 = \frac{81}{36}$ //

v) Variance, $V = \int_{-\infty}^{\infty} x^2 f(x)\,dx - (\mu)^2$

$= \int_{0}^{3} x^2 f(x)\,dx - \left(\frac{81}{36}\right)^2$

$= k \int_{0}^{3} x^4\,dx - \left(\frac{81}{36}\right)^2$ $(\because f(x) = kx^2 \text{ in } (0,3))$

$= \frac{k}{5}\left[x^5\right]_0^3 - \left(\frac{81}{36}\right)^2$

$= \frac{1}{45}\left[3^5\right] - \left(\frac{81}{36}\right)$

$=$

5. Find k such that $f(x) = \begin{cases} kxe^{-x} & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$ is a P.d.f

Find the mean

=> Since $f(x)$ is a P.d.f then

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

$$\int_{-\infty}^{0} f(x)\,dx + \int_{0}^{1} f(x)\,dx + \int_{1}^{\infty} f(x)\,dx = 1$$

$$\int_{0}^{1} kx.e^{-x}\,dx = 1.$$

$$k\int_{0}^{1} x.e^{-x}\,dx = 1$$

$$k\left[\frac{x.e^{-x}}{(-1)} - 1\left(\frac{e^{-x}}{(-1)^2}\right)\right]_0^1 = 1.$$

$$-k\left[e^{-1} + e^{-1} - (0+1)\right] = 1.$$

$$-k\left[\frac{1}{e} + \frac{1}{e} - 1\right] = 1.$$

$$k\left[1 - \frac{2}{e}\right] = 1.$$

$$k = \frac{e}{e-2} \quad \Rightarrow k = \underline{3.785}$$

DATE: PAGE:

Mean, $\mu = \int_{-\infty}^{\infty} x f(x)\,dx$

$= \int_{0}^{1} x\,[k x e^{-x}]\,dx$

$= k \int_{0}^{1} x^2 e^{-x}\,dx$

$= k \left[x^2 \frac{e^{-x}}{-1} - 2x \frac{e^{-x}}{(-1)^2} + 2 \frac{e^{-x}}{(-1)^3} \right]_0^1$

=

6. A c.r.v. x has the P.d.f.

$$p(x) = \begin{cases} kx^2, & -3 \le x \le 3 \\ 0, & \text{otherwise} \end{cases}$$

Evaluate k and f and.

(i) $P[1 < x \le 2]$

(ii) $P[x \le 2]$

iii) $P[x > 1]$

7. A random variable x has the P.d.f $f(x) = \frac{k}{1+x^2}$, $-\infty < x < \infty$. Determine k and hence evaluate

$k = \frac{1}{\pi}$

i) $P(x \ge 0)$

ii) $P[0 < x < 1]$

DATE: PAGE:

Bernoulli's theorem.

The probability of x successors in N trials is equal to ${}^nC_x\, p^x\, q^{n-x}$ where p is the prob of success and q is the prob. of failure.

Suppose a coin is tossed three times and used to know the prob of getting 2 heads. The sample space for tossing a coin 3 times is $S = \{HHH, HHT, HTT, THH, TTH, THT, HTH, TTT\}$. Therefore the prob of getting two heads $= 3/8$. Now we shall obtain the prob. of the event by Bernoulli's theorem. No. of trials, $n = 3$. No. of successes (getting two heads) $= x = 2$. Prob of success, p = prob. of getting a head $= 1/2$. Prob of failure, $q = 1 - p$

$= 1 - 1/2 = 1/2$.

$\therefore {}^nC_x\, p^x\, q^{n-x}$

${}^3C_2\, p^2\, q^{3-2} \Rightarrow 3/8$.

Note :-

Bernoulli's theorem leads to binomial distribution and a limiting case of this distribution leads to poisson distribution. These are discrete probability distribution.

Binomial distribution

If p is the prob. of success and q is the prob. of failure, the prob of x successes out of n trials is given by $P(x) = {}^nC_x\, p^x\, q^{n-x}$. We

DATE: PAGE:

form the following prob distribution of $(x, P(x))$ where $x = 0, 1, 2 \ldots n$

x	0	1	2	n
$P(x)$	${}^nC_0 p^0 q^{n-0}$ q^n	${}^nC_1 p^1 q^{n-1}$	${}^nC_2 p^2 q^{n-2}$	${}^nC_n p^x q^{n-x}$ p^x

Here each $P(x) \geq 0$ and

$$\sum P(x) = {}^nC_0 p^0 q^{n-0} + {}^nC_1 p^1 q^{n-1} + {}^nC_2 p^2 q^{n-2} + \ldots {}^nC_n p^x q^{n-n}$$
$$= [q + p]^n$$
$$= 1^n = 1$$
$$[\because p + q = 1]$$

Hence $P(x)$ is a probability mass function

Mean and S.D of the binomial distribution.

Mean, $\mu = np$

Variance, $V = npq$

S.D, $\sigma = \sqrt{npq}$ or $\sqrt{V}$.

Note

1) The binomial distribution is used for any trial which results in success or failure
2) There are finite no. of trials which are independent.
3) Prob. of success is same in each trial

Problems

DATE: PAGE:

1. Let X be a binomially distributed random variable with mean 2 and S.D $2/\sqrt{3}$. Find the corresponding prob. mass function.

$\Rightarrow$ We have, mean, $\mu = np$

$S.D = \sqrt{v} = \sqrt{npq}$

Since $\mu = 2$ and $\sqrt{v} = \frac{2}{\sqrt{3}}$

$\Rightarrow np = q2$ and $\sqrt{npq} = \frac{2}{\sqrt{3}} \rightarrow (1)$.

$(1) \Rightarrow npq = \frac{4}{3}$ (:squaring)

$2q = \frac{4}{3} \Rightarrow q = \frac{4}{6} = \frac{2}{3}$.

$p = 1 - q$

$= 1 - \frac{2}{3}$

$= \frac{1}{3}$

$np = 2$.

$n\left[\frac{1}{3}\right] = 2$.

$n = 6$.

The p.m.f $P(x) = {}^nC_x\, p^x q^{n-x}$

$= {}^6C_x \left(\frac{1}{3}\right)^x q^{6-x}$

=

$\frac{6 \times 5 \times 4 \times 3}{4 \times 3 \times 2 \times 1}$

DATE: PAGE:

2. Find the prob of getting 4 heads in 6 tosses in a coin

=> The prob of getting a head in single trial

$p = \frac{1}{2}$ => $q = 1 - p$

$1 - \frac{1}{2} = \frac{1}{2}$.

Here $n = 6$, $x = 4$.

We have $P(x) = {}^nC_x \, p^x q^{n-x}$

$= {}^6C_4 \, p^4 \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right)^{6-4}$.

$= 15 \times \frac{1}{16} \times \frac{1}{4}$.

=

3. Determine the binomial distribution for which Mean = 2 (variance) and Mean + variance = 3. Also find prob. of $x \leq 3$. => $P(x \leq 3)$.

=> Given that $\mu = 2v$ and $\mu + v = 3$.

$np = 2npq$. $np + npq = 3$.

$q = \frac{1}{2} = p$. $np[1 + q] = 3$.

$n\left(\frac{1}{2}\right)\left[1 + \frac{1}{2}\right] = 3$

$n = 4$

$\therefore p = \frac{1}{2}$, $q = \frac{1}{2}$, $n = 4$.

$\frac{6 \times 5 \times 4}{3 \times 2 \times 1}$

DATE: PAGE:

$$P(x) = {}^{n}C_x \, p^x \, q^{n-x}$$
$$= {}^{4}C_x \, p^x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{4-x}.$$

$$P(x \leq 3) = 1 - p(x > 3).$$
$$= 1 - P(x = 4)$$
$$= 1 - {}^{4}C_4 \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right)^{4-4}$$
$$= 1 - \frac{1}{16} = \frac{15}{16}.$$

4. Let X be a binomially distributed random variable based on 6 Bernoulli trials, if $p = 0.3$ Evaluate the prob $P(x=3)$, $P(x \leq 3)$, $P(x=4)$ and $P(x>4)$.

=> Given that: $n = 6$ and $p = 0.3$

$$q = 1 - p$$
$$1 - 0.3 = 0.7.$$

$$P(x) = {}^{6}C_x \, (0.3)^x \, (0.7)^{6-x}.$$

$$P(x=3) = {}^{6}C_3 \, (0.3)^3 \, (0.7)^{6-3}$$
$$= 20 \times \quad \times 0.7^3 = 0.18522.$$

$$P(x \leq 3) = 1 - p(x > 3)$$
$$= 1 - [P(x=4) + P(x=5) + P(x=6)]$$
$$= 0.9295$$

DATE: PAGE:

$P(x=4) = {}^6C_4 (0.3)^4 (0.7)^{6-4}$

$= 0.0595$

$P(x>4) = P(x=5) + P(x=6)$

$= 0.0109$

5. If the prob. that a new born child is a male is 0.6. Find the prob. that in a family of 5 children

i) There are exactly 3 boys

ii) Atleast two boys

=> Let x be the male child born.

Given that, $n=5$, $p=0.6$, $q = 1-p$

$1 - 0.6 = 0.4$.

$P(x) = {}^nC_x\, p^x\, q^{n-x}$

$= {}^5C_x (0.6)^x (0.4)^{5-x}$

i) $P(x=3) = {}^5C_3 (0.6)^3 (0.4)^{5-3}$

$= 0.3456$

ii) $P(x \geq 2) = 1 - P(x<2)$

$= 1 - [P(x=0) + P(x=1)]$

$= 1 - [({}^5C_0 (0.6)^0 (0.4)^{5-0}) + ({}^5C_1 (0.6)^1 (0.4)^{5-1})]$

$= 0.913$

6. The number of Telephone lines busy at an instant time is a Binomial variate with prob. 0.2. If at an instant 10 lines are chosen at random, what is the prob. that i) 5

lines are busy
ii) Atmost 2 lines are busy
iii) All lines are busy

=> Let x be the no. of telephone lines busy at an instant time.

Given that : $n = 10$, $p = 0.2$, $q = 1 - p$
$1 - 0.2 = 0.8$

$P(x) = {}^{n}C_x \, p^x \, q^{n-x}$
$= {}^{10}C_x \; 0.2^x \; 0.8^{10-x}$

i) $P(x=5) = {}^{10}C_5 \, (0.2)^5 \, (0.8)^{10-5}$
$=$

ii) $P(x \leq 2) = [P(x=0) + P(x=1) + P(x=2)]$
$=$

$=$

iii) $P(x=10) = {}^{10}C_{10} \, (0.2)^{10} \, (0.8)^{10-10}$
$= 1 \times (0.2)^{10}$
$=$

1. In 256 sets of 12 tosses of a coin, in how many cases one can expect 8 heads and 4 tails

=> The prob. of getting a head, $p = \frac{1}{2}$

$q = 1 - p \Rightarrow 1 - 1/2 \Rightarrow q = 1/2$.

DATE: PAGE:

Given that $n = 12$

$$P(x) = {}^{12}C_x \left[\frac{1}{2}\right]^x \left[\frac{1}{2}\right]^{12-x}$$

By Binomial distribution, the prob of 8 heads and 4 tails in 12 trials is

$$P[x=8] = {}^{12}C_8 \left(\tfrac{1}{2}\right)^8 \left(\tfrac{1}{2}\right)^{12-8}$$
$$= 0.1208$$

∴ The expected no of such cases in 256 sets
$= 256 \times p(x=8) = 30.9 \approx 31$

8 In a quiz contest of answering Yes or No what is the prob. of guessing Yes atleast for 6 answers out of 10 questions asked

9 The prob. of a shooter hitting a target is $1/3$. How many times he should shoot so that the prob of hitting the target atleast one is more than $3/4$.

$\Rightarrow p = \frac{1}{3}$; $q = 1 - p$
$= 1 - 1/3 = 2/3$.

Since $P[x \geq 1] \geq \frac{3}{4}$

$1 - P[x < 1] \geq \frac{3}{4}$

$1 - P(x=0) \geq 3/4$

$$1 - {}^nC_0 \left(\frac{1}{3}\right)^{n-0} \left(\frac{2}{3}\right)^{n-0} \geq \frac{3}{4}.$$

$$1 - \left(\frac{2}{3}\right)^n \geq \frac{3}{4}. \quad -\left(\frac{2}{3}\right)^n \geq \frac{3}{4} - 1 \Rightarrow \left(\frac{2}{3}\right)^n \leq \frac{1}{4}.$$

$n \log 2/3 \leq \log 1/4 \Rightarrow n \leq \log 1/4 / \log 2/3 \Rightarrow 4$

Poisson distibution

The Poisson distribution is regarded as the limiting form of the binomial dist^n when n is very large ($n \to \infty$) and p is very small ($p \to 0$) so that np tends to a fixed finite constant say m i.e., $m = np$ (mean).

The P.D of the random variable is defined as

$$P(x) = \frac{e^{-m} m^x}{x!}$$

$P(x)$ is also called Poisson probability fun^n and x is called a Poisson variate.

The distribution of probabilities for $x = 0, 1, 2, \ldots$ is as follows

x :	0	1	2	.....
$P(x)$:	e^{-m}	$\frac{e^{-1} m}{1}$	$\frac{e^{-2} m^2}{2!}$	.....

We have $P(x) \geq 0$ and $\sum_{x=0}^{\infty} P(x) = \sum_{x=0}^{\infty} \frac{e^{-m} m^x}{x!}$

$$= e^{-m} + e^{-1} m + \frac{e^{-2} m^2}{2!} \ldots$$

$$= e^{-m} \left[1 + \frac{m}{1!} + \frac{m^2}{2!} \ldots\right]$$

DATE: PAGE:

$= e^{-m} \times e^{m} = 1$

$\therefore P(x)$ is a Probability mass function

Mean, $\mu = np = m$.

Variance, $V = m$.

$S.D = \sqrt{V} = \sqrt{m}$.

Note

1) Note that for the poisson distibution, mean = variance = m
2) When p is small and n is large, we approximate binomial distibution by poisson distibutn
3) m is called the parameter of the poisson distibn

Problems

1. If the probability of a bad reaction from a certain injection is 0.001. Determine the chance that out of 2000 individuals more than 2 will get a bad reaction.

=> Given that $n = 2000$; $p = 0.001$

$m = \mu = np$

$= 2000 \times 0.001$

$m = 2$

$\therefore P(x) = \frac{e^{-m} m^{x}}{x!}$

$= \frac{e^{-2} 2^{x}}{x!}$

DATE: PAGE:

Probability that more than 2 will get a bad reaction $= P(x > 2)$

$= 1 - P[x \leq 2]$

$= 1 - [P(x=0) + P(x=1) + P(x=2)]$

$= 1 - \left[\frac{e^{-2} 2^0}{0!} + \frac{e^{-2} 2^1}{1!} + \frac{e^{-2} 2^2}{2!}\right]$

$= 1 - [1 - e^{-2} \times 2 + e^{-2}]$

$=$

DATE: PAGE:

Recall

$$P(x) = \frac{e^{-m} \times m^x}{x!}$$

2. In a certain, turning out razor blades there is a small chance of 0.002 for any blade to be defective. The blades are supplied in packets of 10. Calculate the approximate number of packets containing no defective, 1 defective and 2 defective blades respectively. In a total of 10000 packets

=> Given that $n = 10$ and $p = 0.002$

$\therefore$ Mean, $m = np$

$= 10 \times 0.002 = 0.02$

we have $P(x) = \frac{e^{-m} \times m^x}{x!}$

$= \frac{e^{-0.02} \times x \times (0.02)^x}{x!}$

The probability of no defective blades

$P(x=0) = \frac{e^{-0.02} \cdot 0 \times (0.002)^0}{01}$

$= e^{-0.02} = 0.9802$

Thus the no. of packets containing no defective blades out of 10000 packets =

$10000 \times 0.9802 = 9802$

The probability of one defective blade

$$P(x=1) = \frac{e^{-0.02}(0.02)^1}{1!}$$

$$= 0.019604$$

Thus the number of packets containing a defective blade out of 10000 packets = $10000 \times 0.019604 = 196.04 \approx 196$.

The probability of two defective blades

$$P(x=2) = \frac{e^{-0.02} \times (0.02)^2}{2!}$$

$$= 1.96 \times 10^{-4}$$

Thus the number of packets containing two defective blades out of 10000 packets = $10000 \times 1.96 \times 10^{-4} = 1.96 \approx 2$.

3. If the random variable has a Poisson distribution such that $P(1) = P(2)$, find
   i) Mean of the distribution
   ii) $P(4)$

=> i) Since $P(x) = \frac{e^{-m} m^x}{x!}$

Given that $P(1) = P(2)$

i.e., $P(x=1) = P(x=2)$

$$\frac{e^{-m} m^1}{1!} = \frac{e^{-m} m^2}{2!}$$

DATE: PAGE:

$$m = \frac{m^2}{2}$$

$$m^2 = 2m$$

$$\boxed{m = 2}$$

$$P(x) = \frac{e^{-2} 2^x}{x!}$$

ii) $P(x=4) = \frac{e^{-2} 2^4}{4!}$

=

4. X is a Poisson variable and it is found that the probability that $x=2$ is two thirds of the probability that $x=1$. Find the probability that $x=0$ and probability that $x=3$. What is the probability x exceeds 3

=> Given that $P(x=2) = \frac{2}{3}[P(x=1)]$

Since $P(x) = \frac{e^{-m} m^x}{x!}$

$$\frac{e^{-m} m^2}{2!} = \frac{2}{3}\left[\frac{e^{-m} m^1}{1!}\right]$$

$$\frac{m}{2} = \frac{2}{3} \quad \Rightarrow \quad \boxed{m = \frac{4}{3}}$$

$$P(x) = \frac{e^{-4/3}\left(\frac{4}{3}\right)^x}{x!}$$

DATE: PAGE:

$$P(X=0) = \frac{e^{-4/3}(4/3)^0}{0!}$$

$$= e^{-4/3} = 0.2636$$

$$P(X=3) = \frac{e^{-4/3}(4/3)^3}{3!}$$

$$= 0.1041$$

$$P(X>3) = 1 - P(X \leq 3)$$

$$= 1 - \{P(X=0) + P(X=1) + P(X=2) + P(X=3)\}$$

$$= 0.4645$$

5. A car hire company / firm has two cars which it hires tyres out day by day. The number of demands for a car on each day has the Poisson distribution with mean 1.5. Calculate the proportion of days

i) on which there is no demand

ii) on which some demand is refused

=> Given that $m = 1.5$

Let x denotes the demand for a car on a day. Since $P(x) = \frac{e^{-m} m^x}{x!}$

$$P(x) = \frac{e^{-1.5}(1.5)^m}{x!}$$

i) Proportion of days on which there is no demand = Prob. of there is no demand for the car (i.e., neither of the car used)

$$P(x=0) = \frac{e^{-1.5}(1.5)^{M=0}}{01}$$

$$= 0.2231$$

ii) Proportion of days on which some demand is refused = Prob. that the no. of demands is more than two

$$P(x>2) = 1 - P(x\leq 2)$$

$$= 1 - \{P(x=0) + P(x=1) + P(x=2)\}$$

$$= 1 - \left\{\frac{e^{-1.5}(1.5)^0}{0!} + \frac{e^{-1.5}(1.5)^1}{1!} + \frac{e^{-1.5}(1.5)^2}{2!}\right\}$$

$$= 0.1913$$

6. There is a chance that 5% of the pages of your book contain typographical errors. If 100 pages book is chosen at random, find the probability that two of these pages contain typographical errors using

i) Binomial distibution

ii) Poisson distribution

i) Given that $n = 100$ and $p = \frac{5}{100} = 0.05$

$q = 1 - p$

$1 - 0.05 = 0.95$

we have, $P(x) = {}^nC_x\, p^x\, q^{n-x}$

$= {}^{100}C_x\, (100)^x\, (0.95)^{100-x}$

The prob. that 2 of the pages out of 100 pages contain typographical errors $= P(x=2)$

$P(x=2) = {}^{100}C_2 (0.05)^2 (0.95)^{98}$

$= \underline{0.081}$

ii) Given that $n=100$, $p=0.05$

$m = np$

$= 100 \times 0.05 = 5$

we have $P(x) = \frac{e^{-m} m^x}{x!}$

$= \frac{e^{-5} 5^x}{x!}$

∴ The prob. that 2 of the pages out of 100 pages contain typographical errors $= P(x=2)$

$P(x=2) = \frac{e^{-5} 5^2}{2!}$

$= \underline{0.084}$

Exponential Distribution.

A continuous random variable x is said to follow an exponential distribution with parameter $\alpha > 0$ if P.D.F is given by

$$f = \begin{cases} \alpha e^{-\alpha x} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

$\Rightarrow$ clearly $f(x) \geq 0$ for each $x \geq 0$ and $\alpha > 0$.

$$\int_{-\infty}^{\infty} f(x)\,dx = \int_{-\infty}^{0} f(x)\,dx + \int_{0}^{\infty} f(x)\,dx.$$

DATE: PAGE:

$$= \int_0^\infty \alpha e^{-\alpha x} dx.$$

$$\int_{-\infty}^{\infty} f(x)dx = \alpha \int_0^\infty e^{-\alpha x} dx.$$

$$= \alpha \left[\frac{e^{-\alpha x}}{-\alpha} \right]_0^\infty$$

$$= (-1) \left[e^{-\alpha x} \right]_0^{-\infty}$$

$$= (-1) \left[e^{-\infty} - 1 \right]$$

$$= (-1)[0 - 1] \quad (\because e^{-\infty} = 1/e^{\infty} = 1/\infty = 0)$$

$$= 1$$

$\therefore f(x)$ is a P.D.F

Mean $= \mu = \frac{1}{\alpha}$

Variance $= V = \frac{1}{\alpha^2}$

Standard Deviation $= \sigma = \frac{1}{\alpha} = \sqrt{V}$

Problems

1. If x is an exponential variate with mean 4, Evaluate the following

i) $P[0<x<1]$

ii) $P[x>2]$

iii) $P[-\infty<x<10]$

Given that Mean $= \frac{1}{\alpha} = 4 \Rightarrow \alpha = \frac{1}{4}$

The P.D.F of exponential distribution is

$f(x) = \alpha e^{-\alpha x}, \quad x \geq 0$

$f(x) = \frac{1}{4} e^{-1/4 x}, \quad x \geq 0.$

i) $P[0<x<1] = \int_0^1 f(x)\,dx.$

$= \int_0^1 1/4\, e^{-1/4 x}\, dx.$

$= \frac{1}{4}\left[\frac{e^{-1/4 x}}{(-1/4)}\right]_0^1$

$= (-1)\left[e^{-1/4 x}\right]_0^1$

$= (-1)\left[e^{-1/4} - 1\right]_0^1$

$= \underline{0.2212}$

DATE: PAGE:

ii) $P[x > 2] = \int_2^\infty f(x)\,dx.$

$= \int_2^\infty \frac{1}{4} e^{-1/4 x}\,dx.$

$= \frac{1}{4}\left[\frac{e^{-1/4 x}}{(-1/4)}\right]_2^\infty$

$= (-1)\left[e^{-1/4 x}\right]_2^\infty$

$= (-1)[0 - e^{-1/2}] \quad (\because e^{-\infty} = 1/e^\infty = 1/\infty = 0)$

$= e^{-1/2} = 0.6065$

iii) $P[-\infty < x < 10] = \int_{-\infty}^{10} f(x).\,dx.$

$= \int_{-\infty}^{0} f(x)\,dx + \int_0^{10} f(x)\,dx.$

$= \int_0^{10} \frac{1}{4} e^{-1/4 x}$

$= \frac{1}{4}\left[\frac{e^{-1/4 x}}{-1/4}\right]_0^{10}.$

$= (-1)\left[e^{-1/4 x}\right]_0^{10}$

$= (-1)\left[e^{-5/2} - 1\right] = 0.9179$

DATE: PAGE:

2 In a curtain town, the duration of the shower is exponentially distributed with mean = 5 mins. What is the probability that a shower will last for

i) less than 10 mins
ii) 10 mins or more
iii) between 10 and 12 mins

=> Given that: Mean $= \frac{1}{\alpha} = 5$

$= \alpha = \frac{1}{5}$

The exponential P.D.F is $f(x) = \alpha e^{-\alpha x}$, $x \geq 0$

$f(x) = \frac{1}{5} e^{-1/5 x}$, x

i) $P[x < 10] = \int_{-\infty}^{10} f(x).dx.$

$= \int_{-\infty}^{0} f(x)dx + \int_{0}^{10} f(x).dx.$

$= \int_{0}^{10} \frac{1}{5} e^{-1/5 x} dx.$

$= \frac{1}{5} \left[\frac{e^{-1/5 x}}{-1/5} \right]_0^{10}$

$= (-1)\left[e^{-2} - 1\right] = \underline{0.8647}$

DATE: PAGE:

ii) $P[x \geq 10] = 1 - P[x < 10]$

$= 1 - 0.8647$

$= \underline{0.1353}$

iii) $P[10 < x < 12] = \int_{10}^{12} f(x).dx$

$= \int_{10}^{12} \frac{1}{5} e^{-1/5 x} dx$

$= \frac{1}{5} \left[\frac{e^{-1/5 x}}{(-1/5} \right]_{10}^{12}$

$= (-1) \left[e^{-1/5 x} \right]_{10}^{12}$

$= (-1) \left[e^{-12/5} - e^{-10/5} \right]$

$= \underline{0.0446}$

3 The sales per day in a shop is exponentially distributed with the average sale amounting to Rs 100. and net profit is 8%. Find the probability that the net profit exceeds Rs 30 on two consecutive days.

=> Given that Mean $= \frac{1}{\alpha} = \frac{1}{100} \Rightarrow \alpha = \frac{1}{100}$.

The exponential P.D.F is $f(x) = \alpha e^{-\alpha x}, x \geq 0$.

$f(x) = \frac{1}{100} e^{-1/100 x}, x \geq 0.$

DATE: PAGE:

Let A be the amount for which net profit is 8%

$A \times \frac{8}{100} = 30 \Rightarrow A = 375$

Probability of profit exceeding Rs 30 = $P[x > 30]$

$P[x > 30] = 1 - P[x \leq 30]$

$= 1 - \text{prob}[\text{profit} \leq \text{Rs } 30]$

$= 1 - \text{prob}[\text{sales} \leq \text{Rs } 375]$

$= 1 - \left\{ \int_{-\infty}^{375} f(x).dx \right\}$

$= 1 - \left\{ \int_{-\infty}^{0} f(x).dx + \int_{0}^{375} f(x).dx \right\}$

$= 1 - \int_{0}^{375} \frac{1}{100} e^{-1/100 x} dx.$

$= 1 - \frac{1}{100} \left[\frac{e^{-1/100 x}}{-1/100} \right] dx.$

$= 1 + \left[e^{-1/100 x} \right]_{0}^{375}$

$= 1 + \left[e^{-375/100} - 1 \right] = e^{-375/100}$

$= \underline{0.0235}$

∴ The prob. that profit exceeds Rs 30 on a single day is 0.0235.

Thus the prob. that it reppats on the following day = 0.0235 x 0.0235

= 0.00055

19th = Event on Probability

DATE: PAGE:

Normal Distribution

The continuous probability distribution with P.D.F $f(x)$ is given by $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}$

where $-\infty < x < \infty$, $-\infty < \mu < \infty$ and $\sigma > 0$ is called the normal distribution

Note

1) Mean and S.D of the N.D are μ and σ respectively

Standard Normal Distribution

The continuous random variable x which is distributed normally has the probability fn

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-(x-\mu)^2}{2\sigma^2}}, \quad -\infty < x < \infty$$

$$P[x_1 \le x \le x_2] = \int_{x_1}^{x_2} f(x)\,dx$$

$$= \int_{x_1}^{x_2} \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-(x-\mu)^2}{2\sigma^2}}\,dx$$

let $z = \frac{x-\mu}{\sigma} \Rightarrow \sigma z = x - \mu$

$x = \mu + \sigma z$

$dx = \sigma\,dz$

At $x = x_1$ $z = z_1$ (say)

At $x = x_2$ $z = z_2$ (say)

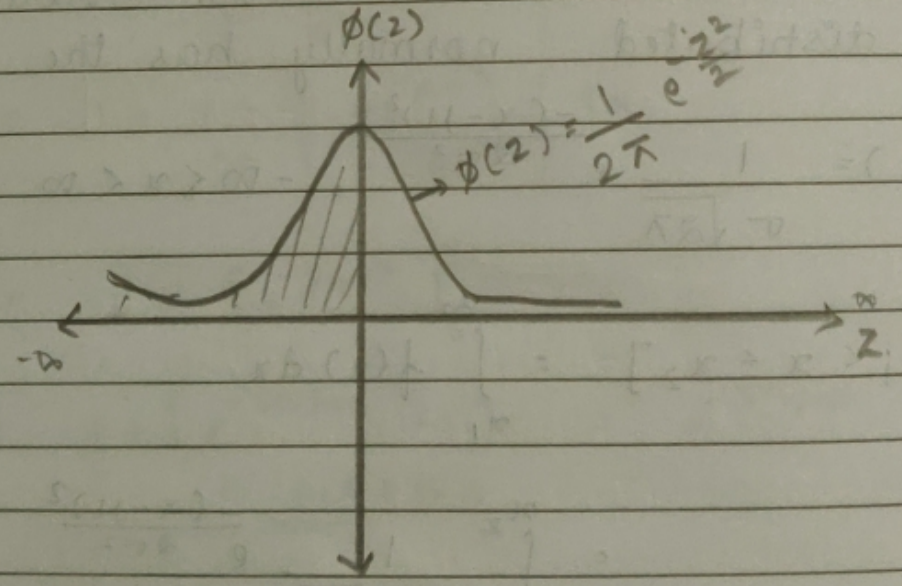
Here $z_1 = \frac{x_1 - \mu}{\sigma}$ and $z_2 = \frac{x_2 - \mu}{\sigma}$

$$P[z_1 \le z \le z_2] = \int_{z_1}^{z_2} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{z^2}{2}} \sigma \, dz.$$

$$= \frac{1}{\sqrt{2\pi}} \int_{z_1}^{z_2} e^{-z^2/2} \, dz$$

$$= \int_{z_1}^{z_2} \phi(z) \, dz.$$

where $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$



The f^n $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$, $-\infty < z < \infty$ is known as the standard N.D where $z = \frac{x - \mu}{\sigma}$

~~Since~~ ~~$\phi(z)$~~ =

DATE: PAGE:

Areas under the normal curve.

i) $\int_{-\infty}^{\infty} \phi(z)\,dz = 1$

ii) $-\int_{z_1}^{0} \phi(z)\,dz = \int_{0}^{z_1} \phi(z)\,dz$

iii) $-\int_{\infty}^{0} \phi(z)\,dz = \int_{0}^{\infty} \phi(z)\,dz = \frac{1}{2}$

iv) $\int_{-\infty}^{z_1} \phi(z)\,dz = \int_{-\infty}^{0} \phi(z)\,dz - \int_{-z_1}^{0} \phi(z)\,dz$

$= 0.5 - \int_{0}^{z} \phi(z)\,dz$

Note:
Equivalently we have i) $P[-\infty \le z \le \infty] = 1$

$= \int_{-\infty}^{\infty} \phi(z)\,dz$

ii) $P[-\infty < z \le 0] = 0.5$

iii) $P[0 \le z \le \infty] = 0.5$ or $P[z \ge 0] = 0.5$

iv) $P[-\infty < z < z_1] = P[-\infty < z < 0] + P[0 < z < z_1]$

$= 0.5 + P[0 < z < z_1]$ or

$P[z < z_1] = 0.5 + P[0 < z < z_1]$

$= 0.5 + \phi(z_1)$

v) $P[z > z_2] = P[0 < z < \infty] - P[0 < z < z_2]$
$= 0.5 - P[0 < z < z_2]$
$= 0.5 - \phi(z_2)$.

Problems

1. Evaluate the following probabilities with the help of Normal Probability table

i) $P[z \geq 0.85]$
ii) $P[-1.64 \leq z \leq -0.88]$
iii) $P[z \leq -2.43]$
iv) $P[|z| \leq 1.94]$

i) $P[z \geq 0.85] = P[0 \leq z < \infty] - P[0 \leq z \leq 0.85]$
$= 0.5 - \phi(0.85)$
$= 0.5 - 0.3023$
$= 0.1977$

ii) $P[-1.64 < z \leq -0.88] = P[0.88 \leq z \leq 1.64]$
$= P[0 \leq z \leq 1.64] - P[0, 0 \leq z \leq 0.88]$
$= \phi(1.64) - \phi(0.88)$
$= 0.4495 - 0.3106$
$= 0.1369$

iii) $P[z \leq -2.43] = P[-\infty < z < 0] - P[-2.43 \leq z \leq 0]$
$= 0.5 - P[0 \leq z \leq 2.43]$ // by symmetry
$= 0.5 - \phi(2.43)$
$= 0.5 - 0.4925$
$= 0.0075$

DATE: PAGE:

iv) $P[|z| \leq 1.94] = P[-1.94 \leq z \leq 1.94]$

$= 2\{P[0 \leq z \leq 1.94]$

$= 2\phi(1.94)$

$= 2 \times 0.4738.$

$= 0.9476.$

2. If z is a normally distributed with mean 0 and variance 1, find

i) $P[z \leq -1.64]$

ii) z_1, if $P[z \geq z_1] = 0.84.$

i) $P[z \leq -1.64] = P[-\infty < z \leq -1.64]$

$= P[1.64 \leq z < \infty]$

$= P[0 \leq z < \infty] - P[0 \leq z \leq 1.64]$

$= 0.5 - \phi(1.64).$

$= 0.5 - 0.4495$

$= 0.0505$

ii) $P[z \geq z_1] = 0.84.$

Since $P[z \geq z_1] > 0.5$; z_1 is -ve.

$P[z_1 \leq z \leq 0] + P[0 \leq z \leq \infty] = 0.84$

$P[0 \leq z \leq z_1] + 0.5 = 0.84.$

$\phi(z_1) = 0.84 - 0.5$

$= 0.34$

Now referring the Normal tables for an area 0.34, we get $z_1 = 0.995$

Since $z_1 = -0.99$

DATE: PAGE:

3. If x is a normally distributed with the mean 12 and S.D 4, find the following

i) $P[x \geq 20]$

ii) $P[x \leq 20]$

=> We have the standard normal variate

$$z = \frac{x - \mu}{\sigma}$$

Given that $\mu = 12$, $\sigma = 4$

$$z = \frac{x - 12}{4}$$

At $x = 20$, $z = \frac{20 - 12}{4} = 2$.

i) $P[x \geq 20] = P[z \geq 2]$

$= P[2 \leq z < \infty]$

$= P[0 \leq z < \infty] - P[0 \leq z \leq 2]$.

$= 0.5 - \phi(2)$

$= 0.5 - 0.4772$

$= 0.0228$.

ii) $P[x \leq 20] = P[z \leq 2]$.

$= P[-\infty < z \leq 2]$

$= P[-\infty < z \leq 0] + P[0 \leq z < 2]$

$= 0.5 + \phi(2)$

$= 0.5 + 0.4772$

$= 0.9772$.

Appendix

Table 1 NORMAL DISTRIBUTION-AREAS

Values of $\Phi(z) = \int_0^z \phi(t)dt$

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	.0000	.0040	.0080	.0120	.01595	.0199	.0239	.0279	.0319	.0359
0.1	.0398	.0438	.0478	.0517	.0557	.0596	.0636	.0675	.0714	.07535
0.2	.0793	.0832	.0871	.0910	.0948	.0987	.1026	.1064	.1103	.1141
0.3	.1179	.1217	.1255	.1293	.1331	.1368	.1406	.1443	.1480	.1517
0.4	.1554	.1591	.1628	.1664	.1700	.1736	.1772	.1808	.1844	.1879
0.5	.1915	.1950	.1985	.2019	.2054	.2088	.2123	.2157	.2190	.2224
0.6	.22575	.2291	.2324	.2357	.2389	.2422	.2454	.2486	.2518	.2549
0.7	.2580	.26115	.2642	.2673	.2704	.2734	.2764	.2794	.2823	.2852
0.8	.2881	.2910	.2939	.2967	.29955	.3023	.3051	.3078	.3106	.3133
0.9	.3159	.3186	.3212	.3238	.3264	.3289	.3315	.3340	.3365	.3389
1.0	.3413	.3438	.3461	.3485	.3508	.3531	.3554	.3577	.3599	.3621
1.1	.3643	.3665	.3686	.3708	.3729	.3749	.3770	.3790	.3810	.3830
1.2	.3849	.3869	.3888	.3907	.3925	.3944	.3962	.3980	.3997	.4015
1.3	.4032	.4049	.4066	.4082	.4099	.4115	.4131	.4147	.4162	.4177
1.4	.4192	.4207	.4222	.4236	.4251	.4265	.4279	.4292	.4306	.4319
1.5	.4332	.4345	.4357	.4370	.4382	.4394	.4406	.4418	.4429	.4441
1.6	.4452	.4463	.4474	.44845	.4495	.4505	.4515	.4525	.4535	.4545
1.7	.4554	.4564	.4573	.4582	.4591	.4599	.4608	.4616	.4625	.4633
1.8	.4641	.4649	.4656	.4664	.4671	.4678	.4686	.4693	.4699	.4706
1.9	.4713	.4719	.4726	.4732	.4738	.4744	.4750	.4756	.4761	.4767
2.0	.4772	.4778	.4783	.4788	.4793	.4798	.4803	.4808	.4812	.4817
2.1	.4821	.4826	.4830	.4834	.4838	.4842	.4846	.4850	.4854	.4857
2.2	.4861	.48645	.4868	.4871	.4875	.4878	.4881	.4884	.4887	.4890
2.3	.4893	.4896	.4898	.4901	.4904	.4906	.4909	.4911	.4913	.4916
2.4	.4918	.4920	.4922	.4925	.4927	.4929	.4931	.4932	.4934	.4936
2.5	.4938	.4940	.4941	.4943	.4945	.4946	.4948	.4949	.4951	.4952
2.6	.4953	.4955	.4956	.4957	.4959	.4960	.4961	.4962	.4963	.4964
2.7	.4965	.4966	.4967	.4968	.4969	.4970	.4971	.4972	.4973	.4974
2.8	.4974	.4975	.4976	.4977	.4977	.4978	.4979	.49795	.4980	.4981
2.9	.4981	.4982	.49825	.4983	.4984	.4984	.4985	.4985	.4986	.4986
3.0	.49865	.4987	.4987	.4988	.4988	.4989	.4989	.4989	.4990	.4990
3.1	.4990	.4991	.4991	.4991	.4992	.4992	.4992	.4992	.4993	.4993
3.2	.4993	.4993	.4994	.4994	.4994	.4994	.4994	.4995	.4995	.4995
3.3	.4995	.4995	.4995	.4996	.4998	.4996	.4996	.4996	.4996	.4997
3.4	.4997	.4997	.4997	.4997	.4997	.4997	.4997	.4997	.49975	.4998
3.5	.4998	.4998	.4998	.4998	.4998	.4998	.4998	.4998	.4998	.4998
3.6	.4998	.4998	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999
3.7	.4989	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999
3.8	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.49995	.49995	.49995
3.9	.49995	.49995	.5000	.5000	.5000	.5000	.5000	.5000	.5000	.5000

For example, (i) $\Phi(1.65) = 0.4505$; (ii) $z = 1.27$ for $\Phi(z) = 0.3980$.